



Firoz's Pizza Express

Research Findings & Business Problem Identification

 Data-Driven Analysis & Operational Insights

 Identifying Key Business Challenges

 Analysis Type

SQL-Based

Data Mining & Pattern Recognition

 Focus Areas

- Runner Performance
- Delivery Operations
- Menu Optimization

 Methodology

4-Step

Systematic Approach

Research Objective & Data Overview

Data Sources Overview



Research Objective

Analyze operational data to identify critical business challenges and uncover opportunities for improvement in delivery operations.



Data-Driven Insights

Transform raw data into actionable findings



Problem Identification

Pinpoint operational bottlenecks



Customer Orders

Pizza selections, toppings, exclusions, and order timestamps



Runner Orders

Delivery assignments, pickup/dropoff times, and distances



Runners

Delivery personnel profiles and registration details



Pizza Recipes

Menu items with standard ingredients and composition



Runners Details

Extended information on runner performance metrics



Ingredients

Inventory data for pizza toppings and additions

Methodology & Data Limitations

Analytical Approach

01

Data Cleaning & Preparation

Standardized formats, handled missing values, and validated data integrity across all tables

02

SQL-Based Analysis

Complex queries to extract metrics, calculate KPIs, and identify performance patterns

03

Pattern Identification

Recognized trends in delivery times, runner behavior, and order characteristics

04

Insight Extraction

Translated data patterns into actionable business insights and problem statements



Data Limitation Notice

-  Analysis is based on 8 orders and 4 runners, hence insights are directional and require validation on larger datasets.
-  Limited sample size reduces statistical significance of findings
-  Patterns observed require further validation with larger datasets before strategic implementation



Methodology Note: All findings derived from systematic SQL-based analysis of transactional data

Runner & Workforce Problems

Problem 1

Inconsistent Runner Performance

Root Cause

Lack of standardized performance tracking and runner scheduling system

Data Insight

Runner 1 delivered 2 orders while Runner 2 delivered 3 orders, observed across 8 total orders

Business Impact

Unreliable service quality and inefficient resource allocation across workforce

Problem 2

Delayed Runner Activation

Root Cause

Inefficient onboarding process and delayed registration verification system

Data Insight

Out of 4 registered runners, only 2 actively completed deliveries in the dataset

Business Impact

Reduced operational capacity with 50% workforce underutilization based on available dataset

Problem 3

Uneven Workload Distribution

Root Cause

Absence of intelligent routing system and manual order assignment process

Data Insight

Active runners handled 2-3 orders each, with no clear distribution pattern observed

Business Impact

Runner burnout risk and potential service quality degradation under uneven workload

Order & Delivery Problems

Problem 1

Extended Delivery Times

Root Cause

Delayed pickup from restaurant due to order preparation bottlenecks

Data Insight

Delivery time ranges from 9 to 22 minutes with 13-minute variation observed across orders

Business Impact

Customer dissatisfaction and competitive disadvantage in market based on dataset

Problem 2

Delayed Order Pickup

Root Cause

Restaurant kitchen preparation time and lack of priority queuing system

Data Insight

Average wait time ~10.8 minutes between order placement and pickup observed in dataset

Business Impact

Delayed delivery start time leading to extended total delivery duration

Problem 3

Order Cancellations

Root Cause

Extended pickup wait times leading to customer-initiated cancellations

Data Insight

1 out of 8 orders was cancelled, observed during high wait time scenarios

Business Impact

12.5% order cancellation rate causing direct revenue loss

Ingredient & Menu Problems

Problem 1

High Ingredient Exclusions

Root Cause

Menu items contain ingredients that customers frequently want removed

Data Insight

3 out of 8 orders contained exclusion requests, observed across multiple pizza types

Business Impact

37.5% of orders require modifications, increasing preparation complexity and time

Problem 2

Frequent Customization Requests

Root Cause

Menu offerings don't align with customer taste preferences

Data Insight

1 out of 8 orders contained extra ingredient requests for customization

Business Impact

Standard menu items don't meet customer needs, indicating menu redesign opportunity

Problem 3

Menu Complexity Issues

Root Cause

Complex menu structure with multiple customization options

Data Insight

50% of orders (4 out of 8) required modifications based on available dataset

Business Impact

Increased order preparation time and potential for errors in custom orders

Summary of Key Business Problems

01



Inconsistent Runner Performance

Out of 4 registered runners, only 2 actively completed deliveries, with 50% workforce underutilization observed across 8 orders

02



Delayed Order Pickup

Average wait time ~10.8 minutes between order placement and pickup, extending total delivery duration

03



Extended Delivery Times

Delivery time ranges from 9 to 22 minutes with 13-minute variation across orders

04



Delayed Runner Activation

Extended gaps between runner registration and first delivery assignment, reducing operational capacity

05



High Ingredient Exclusions

50% of orders (4 out of 8) required modifications, with 37.5% containing ingredient exclusions

06



Order Cancellations

12.5% cancellation rate (1 out of 8 orders) observed during high wait time scenarios

Conclusion



Data-Driven Business Problem Discovery

This comprehensive analysis has successfully identified 6 critical business problems across workforce, delivery, and menu operations through systematic SQL-based data analysis.



Problem Clarity

Transformed raw SQL data into actionable business insights, revealing hidden operational challenges affecting business performance



Root Cause Analysis

Identified underlying causes behind performance issues, enabling targeted understanding of business challenges



Impact Assessment

Quantified business impact of each problem, prioritizing areas requiring immediate attention and strategic focus